

Root Cause Analysis of High Aged AR in Healthcare Billing

Portfolio case study using a public Kaggle healthcare accounts receivable dataset

Prepared for portfolio use

Executive Summary

This project analyzes 200 aged healthcare AR claims to identify the main drivers of unresolved balances and recommend actions a billing organization could take to reduce backlog and improve collections. The dataset contains \$136,623.41 in outstanding balance, with an average unresolved balance of \$683.12 per claim and average aging of 422 days. The two largest root causes are Payer Processing / Follow-up (\$60,643.65, 44.39% of total balance) and Submission / Coding Error (\$55,954.67, 40.96% of total balance). Together, these categories explain roughly 85% of the unresolved AR balance, suggesting the company should focus on both stronger payer escalation and better front-end billing quality controls.

Key KPIs

Metric	Value
Total claims	200
Total billed amount	\$143,364.80
Total outstanding balance	\$136,623.41
Average balance per claim	\$683.12
Average aging days	422
Denied + error share	37.50%
Payer delay balance share	44.39%

Business Problem

A healthcare billing company is carrying a large volume of old unresolved claims in accounts receivable. Leadership wants to understand which issues are driving the backlog, which payers are associated with the highest financial exposure, and what actions should be prioritized first to recover revenue faster.

Objective

The objective of this case study was to identify the primary root causes of unresolved aged AR claims and translate those findings into practical business recommendations that could improve collections and reduce avoidable rework.

Dataset and Tools

This analysis used a public Kaggle dataset containing 200 healthcare AR claims and workflow fields such as status, status code, action code, payer, assigned team, worked by, billed amount, balance amount, and aging days. SQL Server was used to clean, query, and analyze the data. The outputs are suitable for dashboarding in Tableau or Power BI.

Methodology

- Profile the data to measure total outstanding balance, average balance, and aging.
- Group claims by status, status code, payer, assigned team, and action code to see where balance is concentrated.
- Map detailed status codes into broader root-cause categories such as payer processing, submission/coding, eligibility, and clinical/authorization.
- Identify high-risk claims and operational patterns that suggest where billing teams should focus first.

Key Findings

1. Payer processing delays are the largest source of unresolved balance.

The largest root-cause category was Payer Processing / Follow-up with 88 claims and \$60,643.65 in outstanding balance, representing 44.39% of total AR. This indicates that a large share of the backlog is tied to claims still in process, awaiting payer response, or requiring repeated insurance follow-up.

2. Submission and coding defects are nearly as large as payer delay.

Submission / Coding Error accounted for 78 claims and \$55,954.67 in unresolved balance, or 40.96% of total AR. This suggests a significant share of the backlog may be preventable through stronger front-end billing quality controls.

3. Denied, claim-at-insurance, no-response, and pending claims carry the highest exposure.

At the status level, the largest balances were concentrated in Denied, Claim at insurance, No response, and Pending claims. This shows that the problem spans both internal billing quality issues and external payer-side delays.

Status	Claim Count	Total Balance
Denied	49	\$31,625.32
Claim at insurance	46	\$31,590.53
No response	42	\$27,734.21
Pending	37	\$25,246.80
Error	26	\$20,426.55

4. A few issue codes explain most of the balance.

Issue-level analysis showed that a small number of status codes drive most of the unresolved dollars. The largest were Claim in Process, Claim Error, Pending Follow-up, Incorrect Submission, and No Response from Payer.

Status Code	Claim Count	Total Balance
Claim in Process	41	\$28,322.77
Claim Error	26	\$20,426.55
Pending Follow-up	25	\$18,217.28
Incorrect Submission	14	\$10,228.06
No Response from Payer	15	\$9,548.38
Medical Necessity	9	\$7,244.26
Invalid Procedure Code	10	\$7,138.35
Dx inconsistent with CPT	9	\$6,373.59

5. A small group of payers represents the highest financial risk.

The payer view shows that unresolved balance is concentrated in a handful of insurers. This supports payer-specific escalation workflows and standardized follow-up playbooks.

Insurance Name	Claim Count	Total Balance
Medicare	35	\$23,899.65
Cigna	33	\$23,437.91
Aetna	31	\$21,486.73
Community Health Care	30	\$20,596.28
UHC	24	\$16,527.24
Molina	25	\$16,055.27
Humana	22	\$14,620.33

6. Operational effort is heavily focused on rework and follow-up.

The most common and highest-balance action codes were Monitor Claim Status, Need coding assistance, Resubmit claim, Correct & Resubmit, and Follow up with Insurance. This suggests the team is spending substantial effort on rework instead of efficient first-pass resolution.

Action Code	Claim Count	Total Balance
Monitor Claim Status	41	\$28,322.77
Need coding assistance	36	\$26,129.49
Resubmit claim	32	\$21,329.85
Correct & Resubmit	26	\$20,426.55
Follow up with Insurance	25	\$18,217.28
General Follow-up	16	\$9,963.44
Call Insurance	15	\$9,548.38
Post Payment Review	9	\$2,685.65

Business Recommendations

1. Prioritize high-balance denied and error claims first, especially when aging is high. Not all claims should be worked equally; the company should route effort toward the claims with the greatest recovery value and risk.
2. Create payer-specific escalation playbooks for Medicare, Cigna, Aetna, Community Health Care, and UHC, since these payers carry the highest unresolved balance exposure.
3. Strengthen front-end billing quality controls to reduce incorrect submissions, invalid procedure codes, missing modifiers, diagnosis/CPT mismatches, and provider information issues.

4. Separate payer delay work from correction/rework work. Claims stuck with payers require different handling than claims needing coding or submission fixes.
5. Use a risk-based claim prioritization model that combines balance amount, aging days, and status severity rather than relying on aging alone.

Conclusion

The aged AR backlog in this dataset is driven primarily by two issues: payer-side processing delays and internal submission/coding defects. Together, these categories account for roughly 85% of the unresolved balance. A billing organization could use these findings to focus its efforts on high-risk claims, standardize follow-up for major payers, and reduce preventable claim-quality errors upstream. This project demonstrates the ability to translate operational billing data into actionable business recommendations.

Resume-Ready Summary

Analyzed 200 aged healthcare AR claims in SQL Server to identify the main drivers of unresolved balances. Found that payer processing delays and submission/coding errors accounted for approximately 85% of outstanding AR, and recommended risk-based claim prioritization, payer-specific follow-up workflows, and upstream billing quality improvements.